

Database Technologies – Assignment 6

Use Group by clause with appropriate SQL functions to solve following queries. Note: To solve below queries, use “sales” database

1. Write a query that counts the number of salespeople registering orders for each day. (If a salesperson has more than one order on a given day, he or she should be counted only once.).

Select odate, count(distinct snum) from orders group by odate;

```
W2_99546_Pratiksha>Select odate, count(distinct snum) from orders group by odate;
+-----+-----+
| odate      | count(distinct snum) |
+-----+-----+
| 1990-10-03 | 4 |
| 1990-10-04 | 3 |
+-----+-----+
2 rows in set (0.03 sec)
```

2. Write a query on the Customers table that will find the highest rating in each city. Put the output in this form: For the city (city), the highest rating is: (rating).

Select city , max(rating) from customers group by city;

```
W2_99546_Pratiksha>Select city , max(rating) from customers group by city;
+-----+-----+
| city      | max(rating) |
+-----+-----+
| London    | 100 |
| Rome      | 200 |
| San Jose  | 300 |
| Berlin    | 300 |
+-----+-----+
4 rows in set (0.01 sec)
```

- 3 Write an SQL query to calculate the total order amount for each day and display the results in descending order of total orders.

SELECT odate, SUM(amt)

FROM orders

GROUP BY odate

ORDER BY SUM(amt) DESC;

```
W2_99546_Pratiksha>SELECT odate, SUM(amt)
-> FROM orders
-> GROUP BY odate
-> ORDER BY SUM(amt) DESC;
+-----+-----+
| odate      | SUM(amt) |
+-----+-----+
| 1990-10-04 | 16713.81 |
| 1990-10-03 | 8944.59 |
+-----+-----+
2 rows in set (0.00 sec)
```

4. Write a query that selects the total amount in orders for each salesperson for whom this total is greater than the average amount of the order in the table. (Note Use the average amount value directly → 2565.84)

Select snum, sum(amt) as total from orders group by snum having sum(amt) > 2565.84;

```
W2_99546_Pratiksha>Select snum, sum(amt) as total from orders group by snum h
aving sum(amt) > 2565.84;
+-----+-----+
| snum | total |
+-----+-----+
| 1001 | 15382.07 |
| 1002 | 5546.15 |
+-----+-----+
2 rows in set (0.00 sec)
```

5. Write a query that selects the highest rating in each city.

Select city, max(rating) from customers group by city;

```
W2_99546_Pratiksha>Select city, max(rating) from customers group by city;
+-----+-----+
| city | max(rating) |
+-----+-----+
| London | 100 |
| Rome | 200 |
| San Jose | 300 |
| Berlin | 300 |
+-----+-----+
4 rows in set (0.00 sec)
```

6. Retrieve the maximum order amount for each salesperson from the orders table, including only those orders whose value exceeds Rs. 3000.

Select snum, max(amt) from orders group by snum having max(amt) > 3000;

```
W2_99546_Pratiksha>Select snum, max(amt) from orders group by snum having max
(amt) > 3000;
+-----+-----+
| snum | max(amt) |
+-----+-----+
| 1001 | 9891.88 |
| 1002 | 5160.45 |
+-----+-----+
2 rows in set (0.00 sec)
```

7. Write an SQL query to find the smallest order placed by every customer.

Select cnum, min(amt) from orders group by cnum order by cnum asc;

```

W2_99546_Pratiksha>Select cnum,min(amt) from orders group by cnum order by c
num asc;
+-----+-----+
| cnum | min(amt) |
+-----+-----+
| 2001 | 767.19 |
| 2002 | 1713.23 |
| 2003 | 5160.45 |
| 2004 | 75.75 |
| 2006 | 4723.00 |
| 2007 | 1900.10 |
| 2008 | 18.69 |
+-----+-----+
7 rows in set (0.00 sec)

```

Assignment 6 Note:

To solve below queries. use "hr" database

1. Display manager ID and number of employees managed by the manager.
Select manager_id,count(employee_id)from employees group by manager_id;

```

W2_99546_Pratiksha>use hr_db;
Database changed
W2_99546_Pratiksha>Select manager_id,count(employee_id)from employees group by manager_id;
+-----+-----+
| manager_id | count(employee_id) |
+-----+-----+
|          NULL |          1 |
|          100 |          14 |
|          101 |           5 |
|          102 |           1 |
|          103 |           4 |
|          108 |           5 |
|          114 |           5 |
|          120 |           8 |
|          121 |           8 |
|          122 |           8 |
|          123 |           8 |
|          124 |           8 |
|          145 |           6 |
|          146 |           6 |
|          147 |           6 |
|          148 |           6 |
|          149 |           6 |
|          201 |           1 |
|          205 |           1 |
+-----+-----+
19 rows in set (0.00 sec)

```

2. Display the country ID and number of cities we have in the country.

Select country_id,COUNT(city)from locations group by country_id order by country_id asc;

```

W2_99546_Pratiksha>Select country_id,COUNT(city)from locations group by country_id order by country_id asc;
+-----+-----+
| country_id | COUNT(city) |
+-----+-----+
| AU         |          1 |
| BR         |          1 |
| CA         |          2 |
| CH         |          2 |
| CN         |          1 |
| DE         |          1 |
| GB         |          3 |
| IN         |          1 |
| IT         |          2 |
| JP         |          2 |
| MX         |          1 |
| NL         |          1 |
| SG         |          1 |
| US         |          4 |
+-----+-----+
14 rows in set (0.00 sec)

```

3. Display average salary of employees in each department who have commission percentage.

Select department_id,AVG(salary)from employees where commission_pct IS NOT NULL group by department_id;

```

W2_99546_Pratiksha>Select department_id,AVG(salary)from employees where commission_pct IS NOT NULL group by department_id;
+-----+-----+
| department_id | AVG(salary) |
+-----+-----+
|          NULL | 7000.000000 |
|           80 | 8955.882353 |
+-----+-----+
2 rows in set (0.00 sec)

```

4. Display job ID, number of employees, sum of salary, and difference between highest salary and lowest salary of the employees for each job.

Select job_id , COUNT(employee_id),sum(salary),(MAX(salary)-MIN(salary))as difference from employees group by job_id ;

```
W2_99546_Pratiksha>Select job_id , COUNT(employee_id), sum(salary),(MAX(salary)-MIN(salary))as difference from employees group by job_id ;
```

job_id	COUNT(employee_id)	sum(salary)	difference
AC_ACCOUNT	1	8300.00	0.00
AC_MGR	1	12008.00	0.00
AD_ASST	1	4400.00	0.00
AD PRES	1	24000.00	0.00
AD_VP	2	34000.00	0.00
FI_ACCOUNT	5	39600.00	2100.00
FI_MGR	1	12008.00	0.00
HR_REP	1	6500.00	0.00
IT_PROG	5	28800.00	4800.00
MK_MAN	1	13000.00	0.00
MK_REP	1	6000.00	0.00
PR_REP	1	10000.00	0.00
PU_CLERK	5	13900.00	600.00
PU_MAN	1	11000.00	0.00
SA_MAN	5	61000.00	3500.00
SA_REP	30	250500.00	5400.00
SH_CLERK	20	64300.00	1700.00
ST_CLERK	20	55700.00	1500.00
ST_MAN	5	36400.00	2400.00

19 rows in set (0.00 sec)

5. Display job ID for jobs with average salary more than 10000.

Select job_id,AVG(salary)from employees group by job_id having AVG(salary)>10000;

```
W2_99546_Pratiksha>Select job_id,AVG(salary)from employees group by job_id having AVG(salary)>10000;
```

job_id	AVG(salary)
AC_MGR	12008.000000
AD PRES	24000.000000
AD_VP	17000.000000
FI_MGR	12008.000000
MK_MAN	13000.000000
PU_MAN	11000.000000
SA_MAN	12200.000000

7 rows in set (0.00 sec)

6. Display the years in which more than 10 employees joined.

Select year(hire_date), COUNT(employee_id)

from employees

GROUP BY YEAR(hire_date)

having COUNT(employee_id)>10;

```
W2_99546_Pratiksha>SELECT YEAR(hire_date), COUNT(employee_id)
-> FROM employees
-> GROUP BY YEAR(hire_date)
-> HAVING COUNT(employee_id) > 10;
```

YEAR(hire_date)	COUNT(employee_id)
2015	29
2016	24
2017	19
2018	11

4 rows in set (0.02 sec)

7. Display departments in which more than five employees have commission percentage.

SELECT department_id,COUNT(employee_id)from employees where commission_pct IS NOT NULL group by department_id having COUNT(employee_id)<5;

```
W2_99546_Pratiksha>SELECT department_id,COUNT(employee_id)from employees where commission_pct IS NOT NULL group by department_id having COUNT(employee_id)<5;
```

department_id	COUNT(employee_id)
NULL	1

1 row in set (0.00 sec)

8. Display employee ID for the employees who did more than one job in the past.

Select employee_id,count(job_id)from job_history group by employee_id having count(job_id)>1;

```
W2_99546_Pratiksha>Select employee_id,count(job_id)from job_history group by employee_id having count(job_id)>1;
```

employee_id	count(job_id)
101	2
176	2
200	2

3 rows in set (0.00 sec)

9. Display job ID of jobs that were done by 2 or more employees for more than 100 days individually.

Select job_id , count(employee_id) from job_history group by job_id having count(employee_id)>=2;

```
W2_99546_Pratiksha>Select job_id , count(employee_id) from job_history group by job_id having count(employee_id)>=2;
```

job_id	count(employee_id)
AC_ACCOUNT	2
ST_CLERK	2

2 rows in set (0.00 sec)

10. Display the department ID, year of hiring, and the number of employees who joined each department in each year.

Select department_id , year(hire_date),count(employee_id)as employees from employees group by department_id , year(hire_date) order by department_id;

```
W2_99546_Pratiksha>Select department_id , year(hire_date),count(employee_id)as employees from employees group by department_id , year(hire_date) order by department_id;
```

department_id	year(hire_date)	employees
NULL	2017	1
10	2013	1
20	2014	1
20	2015	1
30	2012	1
30	2013	1
30	2015	2
30	2016	1
30	2017	1
40	2012	1
50	2013	3
50	2014	4
50	2015	12
50	2016	13
50	2017	9
50	2018	4
60	2015	1
60	2016	2
60	2017	2
70	2012	1
80	2014	5
80	2015	10
80	2016	7
80	2017	5
80	2018	7
90	2011	1
90	2013	1
90	2015	1
100	2012	2
100	2015	2
100	2016	1
100	2017	1
110	2012	2

33 rows in set (0.00 sec)

11. Display the count of employees joined in each month of the year 2017.

SELECT count(employee_id), monthname(hire_date)from employees where year(hire_date)=2017 group by monthname(hire_date);

```
W2_99546_Pratiksha>SELECT count(employee_id), monthname(hire_date)from employees where year(hire_date)=2017 group by monthname(hire_date);
```

count(employee_id)	monthname(hire_date)
2	May
3	February
3	December
1	August
2	November
1	January
1	April
1	October
3	March
2	June

10 rows in set (0.00 sec)

12. Display the details of departments in which the maximum salary is more than 10000.

SELECT department_id,MAX(salary)from employees group by department_id having MAX(salary)>10000;

```
W2_99546_Pratiksha>SELECT department_id,MAX(salary)from employees group by department_id having MAX(salary)>10000;
+-----+-----+
| department_id | MAX(salary) |
+-----+-----+
|          20  | 13000.00   |
|          30  | 11000.00   |
|          80  | 14000.00   |
|          90  | 24000.00   |
|         100  | 12008.00   |
|         110  | 12008.00   |
+-----+-----+
6 rows in set (0.02 sec)
```